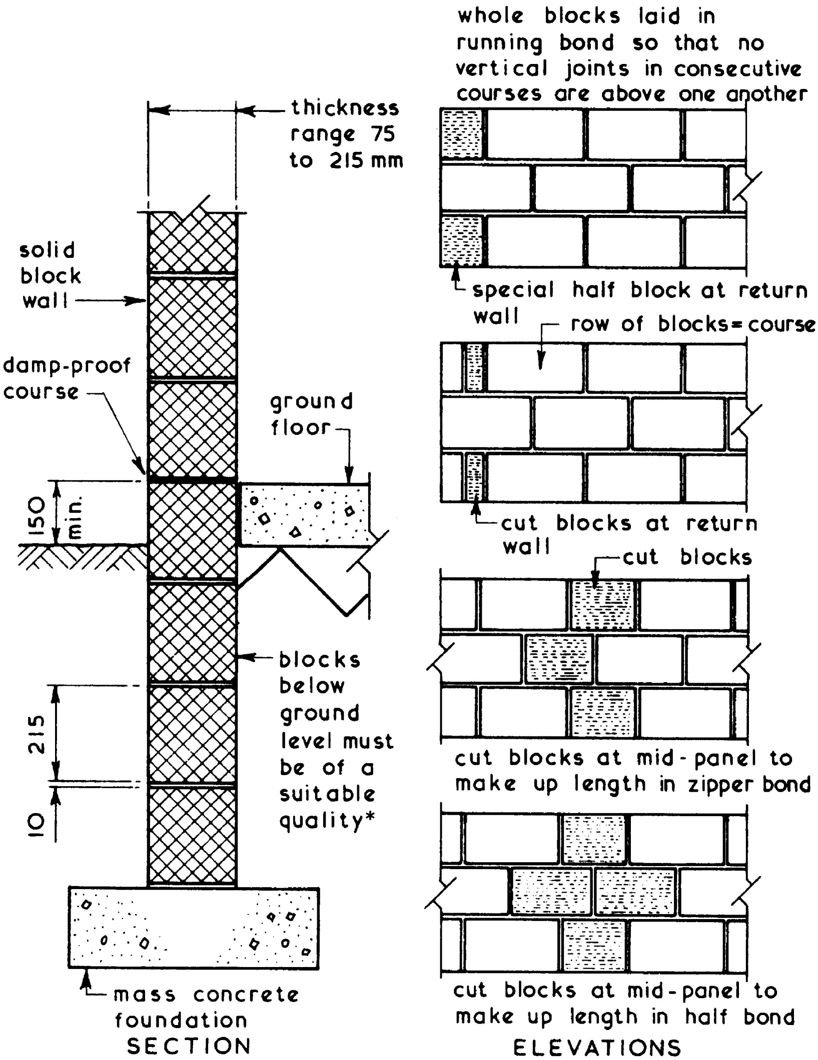
Blocks ~ these are walling units exceeding in length, width or height the dimensions specified for bricks in BS EN 772-16. Precast concrete blocks should comply with the recommendations set out in BS 6073-2 and BS EN 771-3. Blocks suitable for external solid walls are classified as load bearing and are required to have a minimum declared compressive strength of not less than 2.9 N/mm^2.

Typical Details ~

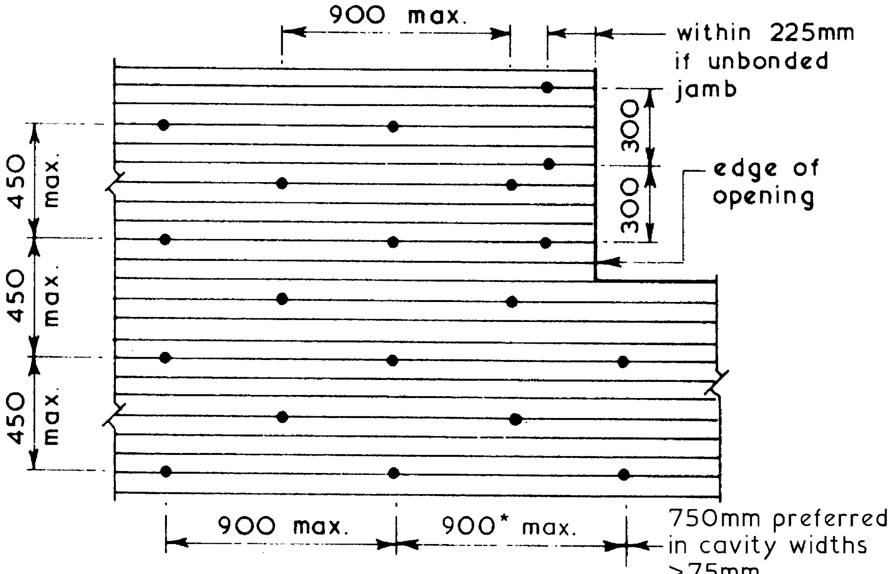


*See pages 410 to 412.

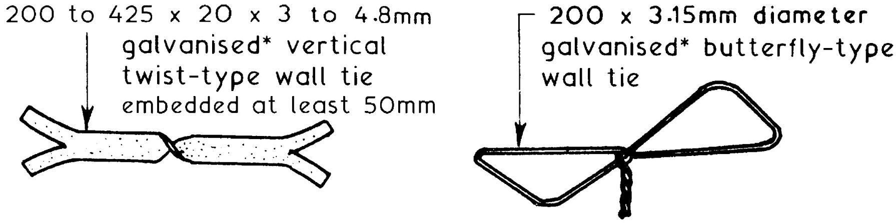
Refs.: BS 6073-2: Precast concrete masonry units.
BS EN 772-16: Methods of test for masonry units.
BS EN 771-3: Specification for (concrete) masonry units.

Cavity Walls ~ these consist of an outer brick or block leaf or skin separated from an inner brick or block leaf or skin by an air space called a cavity. These walls have better thermal insulation and weather-resistance properties than a comparable solid brick or block wall and therefore are in general use for the enclosing walls of domestic buildings. The two leaves of a cavity wall are tied together with wall ties located at 2.5/m^2, or at equivalent spacings shown below and as given in Section 2C of Approved Document A – Building Regulations.

With butterfly-type ties the width of the cavity should be between 50 and 75 mm. Where vertical twist-type ties are used the cavity width can be between 75 and 300 mm. Cavities are not normally ventilated and are closed by roof insulation at eaves level.



WALL TIE SPACINGS



TYPICAL METAL WALL TIES

Ref: BS EN 845-1: Specification for ancillary components for masonry. Wall ties, tension straps, hangers and brackets.

* Stainless steel or non-ferrous ties are now preferred.

Minimum requirements ~
Thickness of each leaf, 90 mm.
Width of cavity, 50 mm.
Wall ties at 2.5/m^2 (see previous page).
Compressive strength of bricks, 6 N/mm^2 up to two storeys.*
Compressive strength of blocks, 2.9 N/mm^2 up to two storeys.*

* For work between the foundation and the surface a 6 N/mm^2 minimum brick strength is normally specified or 2.9 N/mm^2 minimum concrete block strength (see next page for variations).

Combined thickness of each leaf + 10 mm whether used as an external wall, a separating wall or a compartment wall, should be not less than 1/16 of the storey height** which contains the wall.

** Generally measured between the undersides of lateral supports, e.g. undersides of floor or ceiling joists, or from the underside of upper floor joists to half-way up a laterally restrained gable wall. See Approved Document A, Section 2C for variations and next page.

Wall dimensions for minimum combined leaf thicknesses of 90 mm + 90 mm + 10 mm (190 mm min. actual thickness) ~

Height	Length
3.5 m max.	12.0 m max.
3.5 m – 9.0 m	9.0 m max.

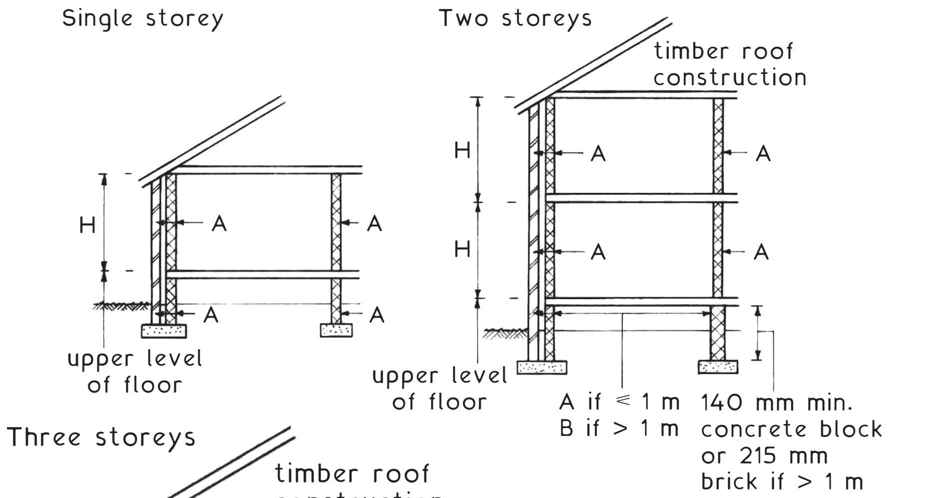
Wall dimensions for minimum combined leaf thickness of 280 mm + 10 mm (290 mm actual thickness), e.g. 190 mm + 100 mm for one storey height and a minimum 180 mm combined leaf thickness + 10 mm (i.e. 90 mm + 100 mm) for the remainder of its height ~

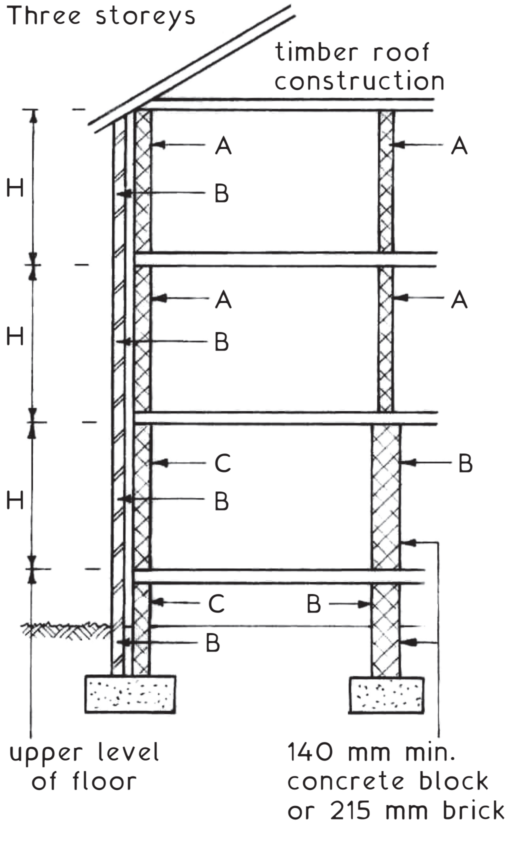
Height	Length
3.5 – 9.0 m	9.0 – 12.0 m
9.0 m – 12.0 m	9.0 m max.

Wall dimensions for minimum combined leaf thickness of 280 mm + 10 mm (290 mm actual thickness) for two-storey height and a minimum 180 mm combined leaf thickness + 10 mm (190 mm actual thickness) for the remainder of its height ~

Height	Length
9.0 m – 12.0 m	9.0 m – 12.0 m

Wall length is measured from centre to centre of restraints by buttress walls, piers or chimneys.
For other wall applications, see the reference to calculated brickwork on page 431.

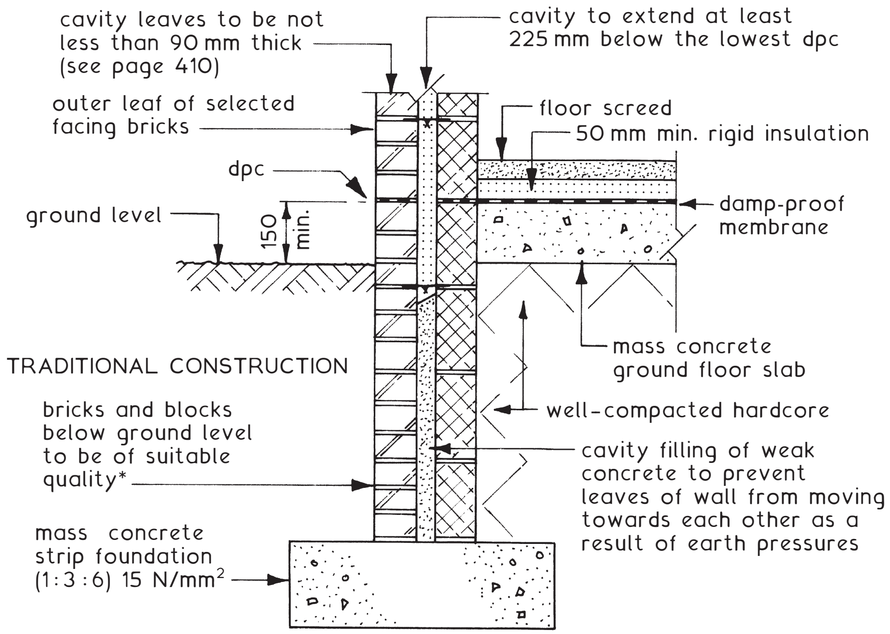


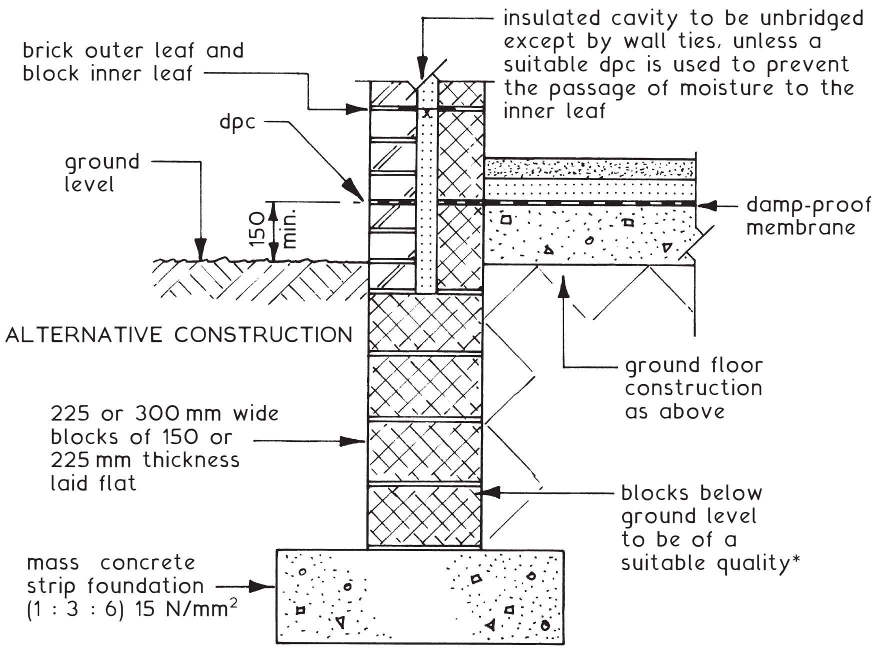


Key for H ≤ 2.7 m (N/mm^2 min. compressive strength) ~

	Conc.blocks	Bricks
A	2.9	6.0
B	7.3	9.0
C	7.3	18.0

Note 1: If H > 2.7 m, A should be at least 7.3 N/mm^2 for concrete blocks and 9 N/mm^2 for brickwork, or as indicated in the diagrams (take greater value).
Note 2: The above applies to standard format bricks of 215 mm × 102.5 mm × 65 mm with < 25% formed voids (20% for bricks with frogs). For bricks with voids > 25% (max. 55%) see Table 6 in Approved Document A, Section 2C.





*Min. compressive strength depends on building height and loading. See Building Regulations A.D. A: Section 2C (Diagram 9) and extracts shown on the previous page.